nanoker

CERAMIC TECHNICAL DATA SHEET

NC2

Ce-stabilized ATZ Nanocomposite

KEY PROPERTIES

- Ultra-high toughness
- High strength
- Elastic-plastic behavior
- No ageing (no low temperature degradation)
- High wear resistance under sliding conditions
- Low friction coefficient against metals

APPLICATIONS

- Structural parts for machinery
- Welding pins
- Drilling tools
- Cutting blades

MAIN PROPERTIES

Property	Value	Unit
Density	5.40	g/cm³
Flexural strength	900	MPa
Fracture Toughness	10.0	MPa·√m
Hardness	10.5	GPa
Young Modulus	245	GPa
Thermal Conductivity	10	W/m·K
CTE (20–1000ºC)	9.3	$x10^{-6}$ K^{-1}
Weibull Modulus	18	-
Electrical Resistivity	$>10^{11}$	Ω·cm
Electrical Behavior	Insulator	-

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1200*	ºC
Recomended temp	<1100	ºC
Continuous temp	1000	ºC

* (estimado típico ATZ)

CHEMICAL COMPOSITION

$ZrO_2 + HfO_2$	6	0 – 70 %
CeO_2		8 – 13 %
Al_2O_3		20 – 30 %
Other oxides		< 0.5 %

MATERIAL INFORMATION

Material type	Ce-stabilized ATZ nanocomposite
Structure	Nanocomposite ceramic
Key feature	Ultra-high toughness
Machinability	Medium
Color	White

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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